Common section

soon discovered, however, that being a vegetarian wasn't all that easy in a society of carnivores. Her brother, Bart, tormented her mercilessly. Lisa faced another crisis when her father, Homer, hosted a pig roast. When Lisa's offer of vegetarian gazpacho as an alternative was rejected, she destroyed the barbecue and fled in disgust at her family's indifference to animal suffering.

Lisa found solace and moral support from her friends the McCartneys, who were visiting Apu, a vegan from India. Lisa confided in Apu that she eats cheese. "You must think I'm a monster," Lisa said. "Indeed I do," Apu responded. "But I learned long ago to tolerate others rather than forcing my beliefs on them." Lisa realized that one cannot convert others through force but only through reason and example.

THE LEGAL AND MORAL STATUS OF NONHUMAN ANIMALS

Most people, when asked, say that they like animals and disapprove of animal cruelty. They draw the line, however, when it comes to animals having rights. When there is a conflict between animal interests and human interests, no matter how trivial, human interests almost always win out. We eat meat, visit zoos, wear leather shoes, use cosmetics and drugs that have been tested on captive animals, and abandon our dogs and cats at animal shelters when they inconvenience us.

Suggestions that animals have rights that we ought to respect are generally met with ridicule. When Mary Wollstonecraft published *A Vindication of the Rights of Woman* in 1792, her ideas were also ridiculed. One critic wrote that if the idea of equality was applied to women, why shouldn't it also hold for "brutes"? Because the idea of granting rights to other animals is absurd, the critic concluded, it is also absurd to grant rights to women. Women have since been granted rights under the law, although it took more than a century.

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In the past four decades, dozens of books and hundreds of articles have been written on animal rights. Thousands of animal-rights and vegetarian groups have proliferated; most colleges have at least one animal-rights group on campus. Much of the inspiration for the current animal-rights movement has come from Eastern philosophy. Gandhi's philosophy, in particular, has had an enormous influence on both the American civil rights movement and the animal-rights movement.

In many Asian countries, such as India, vegetarianism has long been the norm. According to Hindu and Buddhist ethics, meat-eating violates the principle of *ahimsa*, or nonviolence. Meat-eating also contributes to a mentality of violence and has negative karmic consequences. Japan has banned research using great apes (chimpanzees, gorillas, bonobos, and orangutans); in other Asian countries, apes have never been used in scientific research.

Among Western nations, the Spanish parliament in 2008 approved a resolution that grants great apes "human rights," including the right to life and protection from harmful research practices and commercial exploitation. Other countries, including the Netherlands, Britain, Austria, Switzerland, Sweden, Australia, and New Zealand, have bans on research using great apes. There is pressure in some of these countries to bestow legal "personhood" on great apes. In the United States, which uses more chimps in biomedical research than any other country, there is greater resistance to the

The Animal Welfare Act was enacted in the United States in 1970. The act is not concerned, however, with protecting the welfare of rats, mice, birds, reptiles, frogs, or animals raised for food; nor does it include genetically altered animals. In addition, animal welfare laws do not give animals rights. Concern for animal suffering is based primarily on the concept of "necessary suffering." Suffering is wrong ("unnecessary") only when it does not advance human interests.

Ethical positions regarding public policy can be divided into the abolitionist, reformist, and status quo positions. *Abolitionists* argue that we should stop using animals altogether as a source of food and as tools in scientific experiments. They oppose zoos, circuses, and keeping animals as house pets. (See Case Study 4: *Zoos: Prisons or Havens*.) Their position is supported by animal-rights activists such as Tom Regan. (For more on Regan's position on animal rights, see reading by Cohen at the end of this chapter.) Although few Western philosophers accept this position, it is more generally accepted in Eastern philosophies such as Hinduism, Buddhism, Taoism, and Jainism, which are based on respect for all living beings.

Reformists accept meat-eating and animal experimentation, but think that we need to improve these institutions. Unlike abolitionists, who sometimes work outside the law, reformist organizations such as humane societies work within the system, promoting reform and legislative initiatives on behalf of animals. Animal-welfare laws that require cages of a certain size for animals in laboratories but do not oppose animal experimentation itself are based on the reformist position.

The *status quo* position maintains that no changes are necessary in the way humans treat other animals. The status quo position is supported by traditional Western philosophies, which regard nonhuman animals as lacking moral value. This philosophical tradition has been one of415the most deeply rooted obstacles to any serious consideration of the moral rights of both nonhuman animals and the environment.

THE LEGAL AND MORAL STATUS OF THE ENVIRONMENT

Environmental ethics is concerned with the moral basis of environmental responsibility, including the moral value of nonhuman nature, pollution, population control, food production, and preservation of wilderness and species diversity.

The environmental ethics movement is a relative newcomer on the philosophical scene. In 1967 history professor Lynn White published an article in *Science* in which he blamed Christian anthropocentric thinking for creating an ecological crisis. According to Judeo-Christian tradition, the earth was created by God for the benefit of humans. "By destroying pagan animism," White wrote, "Christianity made it possible to exploit nature in a mood of indifference to the feelings of natural objects."[3]

Philosophers entered the fray in the early 1970s. Their attention was initially focused on Aldo Leopold's landmark essay on land ethics, excerpts of which are included in this chapter. Leopold attacks the anthropocentric relationship between humans and the environment that is based on humans as conquerors of the land. He calls on us to replace the old paradigm with a new ecocentric paradigm in which humans are viewed as members, rather than conquerors, of the greater biotic

As in the animal-rights debate, the *status quo* position is based on an anthropocentric worldview that regards nonhuman nature as a resource for human consumption. The primary purpose for protecting nonhuman nature, according to this view, is to preserve it for ourselves and future generations. The growing awareness of global warming and the detrimental effects of certain human activities has led to greater concern for protection of the environment.

THE PHILOSOPHERS ON THE MORAL VALUE OF NONHUMAN ANIMALS AND THE ENVIRONMENT

Ancient Greek philosophers were divided regarding the moral value of nonhuman nature. The Pythagoreans, who were vegetarians, taught that other animals should be treated with respect. Aristotle, on the other hand, thought that other animals had only instrumental moral value. Contemporary views on the moral value of nonhuman animals have been shaped primarily by Aristotle's anthropocentric philosophy.

The split between humans and nonhuman nature in Western thought began with the acceptance of the ancient Greek dualism, which split reality into nonthinking material substances416(body) and nonmaterial thinking substances (mind or soul). Aristotle argued that reason is an activity of the soul. Humans have a soul; nonhuman animals don't. Because moral value depends on the ability to reason, only humans have intrinsic moral value. Other animals are inferior beings whom nature has made "for the sake of man."[4]

Aristotle's worldview was Christianized by Thomas Aquinas. The world was created by God for humans. "Humans," according to Aquinas, "are the highest in the order of material beings, yet the lowest in the order of spiritual beings … the progression from the non-living to humans is one of increasing perfection … schematically, humans are at the apex of material creation."[5] According to Genesis 1:26, "God said, 'let us make man in our image, after our likeness; and let them have dominion over the fish of the sea, and over the birds of the air, and over the cattle … and over all the earth'." Nature and other animals exist only to the extent that they benefit humans.

The belief that humans are separate from, and morally superior to, other animals was affirmed by René Descartes, the "father of modern philosophy." Carrying the hierarchical worldview of his predecessors even further, Descartes concluded that nonhuman animals are merely organic machines, much like clocks, without souls, free will, or consciousness. Because of this, it is not immoral to kill and eat them.

John Locke based his natural rights theory on the belief that humans are a special and unique creation of God. To base rights on equal consideration for the interests of all living beings, rather than on the so-called special nature of humans, is to deny the "natural" order of creation. Locke's anthropocentric, theologically based worldview granted humans the inalienable right to exploit nonhuman nature with impunity.

Francis Bacon (1561–1626), founder of the scientific method, unquestionably accepted the prevailing philosophical view on nonhuman animals. Bacon enthusiastically advocated **vivisection**—the dissection of live animals—for the pure joy of learning. Because of the tremendous success of science in generating results and new technologies, few people bothered to question the morality of

human sensibilities. On the other hand, because "animals must be regarded as man's instruments," we can use them for scientific experiments and for food just as we can use the natural environment for justifiable human purposes.

The battle for equal rights for all groups of humans fueled a similar demand for respect for the rights of other animals as well. Mohandas Gandhi, Mary Wollstonecraft, Susan B. Anthony, and Elizabeth Cady Stanton, to name only a few human-rights advocates, also spoke out on behalf of other animals. The utilitarians were among the first advocates for nonhuman animals. According to Jeremy Bentham, it is not reason but the capacity to suffer that is morally relevant.[6] The utilitarian's concern for the happiness of all sentient beings—regardless of their race, gender, or species—reflects the moral ideal of equality that was so important during the late eighteenth century. This ideal gave rise to both the American and French Revolutions. The utilitarians, such as Bentham, hoped that this moral ideal would someday be extended to all sentient beings.

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Charles Darwin (1809–1882) also rejected anthropocentricism. Darwin attacked the assumption that only humans are capable of reason. Reason, by definition, involves the ability to form general rules from particular experiences. That other animals are capable of reasoning seemed obvious to Darwin. "Only a few persons now dispute that animals possess some power of reasoning," he wrote in *The Descent of Man* (1871). "Animals may constantly be seen to pause, deliberate, and resolve."[7] But Darwin underestimated the power of tradition. His theory of evolution was reinterpreted not only to justify the oppression of non-Western people, but to justify the exploitation of other animals and nature by placing humans at the apex of evolution, thus scientifically legitimating the religious view that humans are a special creation.

Like Darwin, primatologist Frans de Waal points out how very similar the emotional and moral behavior of humans is to that of other animals.[8] Many social animals demonstrate what seems to be a natural moral sense and will respond to the distress of another animal, even one of another species. Not only are many of these animals capable of reason, but they also appear to deliberate and interact with one another according to a sense of fairness and reciprocity.

Feminist philosophers are divided on the issue of animal rights. Many, if not more, argue that we do not have moral obligations toward nonhuman animals because they are incapable of being in reciprocal caring relationships with humans. Other feminists, such as Karen Warren, argue that the domination of women and the domination of nonhuman animals are part of the same patriarchal paradigm. Women can achieve autonomy only by rejecting the dualistic ideology that allows humans to subordinate other animals.

ANIMAL FARMING, ENVIRONMENTAL DEGRADATION, AND VEGETARIANISM

Our eating habits are our most direct interaction with the earth. Americans are steeped in the Jeffersonian tradition that regards farmers as "the chosen People of God" and our "most valued citizens."[9] Farming, Jefferson claimed, ennobles humans by keeping them in touch with living nature. However, modern agriculture is not a natural process but a cultural institution shaped by

The affluence boom that followed World War II was accompanied by an increase in meat-eating, which was regarded as a status symbol. By the early 1970s, rich nations were feeding more grain to their livestock than all the people of China and India (who make up more than two-thirds of the world's population) consumed directly.

Intensive farming of animals also began in the United States shortly after the war. Intensive farming involves raising animals indoors in large, automated "factories." Several thousand chickens or pigs may be housed in one building. Confining animals in buildings requires less land and less labor. The modern factory farm reflects the traditional Western view of nonhuman animals as418machines. In line with this, the Food and Drug Administration (FDA) has recently moved toward approving the creation of genetically engineered animals for meat production.

The institution of animal agriculture is one of the greatest sources of suffering for nonhuman animals. Animals in the modern factory farm are raised in large buildings in crowded cages or stalls. Today almost all egg production comes from caged birds in automated factory buildings.[11] Journalist Joy Williams writes:

Factory farmers are all Cartesians. Animals are no more than machines— milk machines, piglet machines, egg machines—production units converting themselves into profits…. The factory farm today is a crowded, stinking bedlam, filled with suffering animals that are quite literally insane, sprayed with pesticides and fattened on a diet of growth stimulants, antibiotics, and drugs. Two hundred and fifty thousand laying hens are confined within a single building. (The high mortality rate caused by overcrowding is economically acceptable; nothing is more worthless than an individual chicken.)[12]

We need to eat to stay alive. We do have a choice, however, in what to eat. Humans do not need meat in their diets. Indeed, heavy reliance on meat is one of the leading causes of disease and obesity.[13]

More and more people in countries such as the United States, Canada, England, and Australia are becoming vegetarians. A **vegetarian** is a person who refrains from eating meat, poultry, and seafood. **Vegans** abstain from all animal products, including eggs and milk. In a 2016 Harris poll, 3.3 percent of the people responded that they were vegetarians, more than double the number from a 1977 survey.[14]

Ethical vegetarians maintain that animals have moral value. Although meat-eating might be justified if we needed meat for survival, the human taste for animal flesh does not justify killing and eating animals. Those who defend meat-eating deny that animals have either moral status or rights that we must respect.

Environmental ethicists do not have a unified position on agriculture. Rather than rejecting it, many judge it against its impact on the entire biotic community. While many environmental ethicists advocate for more humane treatment of nonhuman animals, a few support a move toward intensive farming of "farm" animals, blaming grazing "livestock" for the destruction of wildlife habitat. This has led to criticism that environmentalists who identify nature with the pristine wilderness have

much fossil fuel as a vegetarian diet. One acre of land can produce 40,000 pounds of potatoes or 250 pounds of beef.[15] Much of the water pollution, depletion of topsoil, and deforestation is the result of animal agriculture. In the United States, most of the agricultural land is dedicated to raising beef.

Solid waste disposal also contributes to environmental degradation. While population growth in the Third World has been blamed for many of the environmental problems we are currently419facing, in fact, affluent Westerners are responsible for much of this pollution. The average American produces much more household waste and is responsible for more industrial and agricultural pollution than anyone else in the world. Hazardous waste disposal, including the disposal of nuclear waste, is particularly problematic. War also has a negative effect on the environment.

The environmental justice movement emerged in the United States in the 1980s. Its focus has been on the urban environment rather than on the wilderness. As such, this movement has radically redefined the meaning of environment to include the broader framework of economic, racial, and social justice. The movement's primary concerns have been ending environmental racism and promoting tighter government regulation of industrial pollution and waste disposal. The BP Gulf Oil spill in 2010 and approval of oil drilling in the Arctic has drawn the attention of environmentalists who, in general, oppose oil drilling in pristine wilderness areas. (See Case Study 7: *The Gulf Oil Spill Disaster*.)

GLOBAL WARMING AND CLIMATE CHANGE

Back in the sixteenth century, when William Shakespeare took in a breath, 280 molecules out of every million entering his lungs were carbon dioxide. Today, each time you breathe in, 380 molecules out of every million are carbon dioxide (CO_2).[16] Most scientists believe that the increase in CO_2 is primarily due to human activities and that emissions of CO_2 and other greenhouse gases are the leading cause of a phenomenon known as **global warming**.

Global warming and climate change are of concern to environmental ethicists because of the impact of our consumer-oriented lifestyles, especially the use of CO_2 emitting fossil fuels, on global warming. Global warming has been accelerating in the past decade at a faster than predicted rate. According to NASA, the six hottest years on record have been since 2010, with 2017 being the second hottest year in recorded history.[17]

As a result of global warming, the polar ice caps and mountain glaciers are melting, storm activity is increasing, precipitation patterns are changing, and oceans are warming, leading to rising sea levels and coastal flooding. It is predicted that disruption of the ecosystem by global warming will lead to massive species extinction, with 15 to 35 percent of species being at risk by 2050, a trend that has already begun with the destruction of rain forests for lumber and agriculture. (See Case Study 6: *Earth's Dwindling Forests*.)

The United States is the single largest emitter of greenhouse gases. The Kyoto Protocol, which took effect in 2005, assigned the 169 signatory nations mandatory targets for reduction in emissions of greenhouse gases. Although the United States signed the treaty, it refuses to ratify it because of the

Millions of nonhuman animals are killed every year in scientific experiments in the United States alone. As with animal agriculture, the practice of animal experimentation is based on an anthropocentric paradigm.

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The U.S. Food and Drug Administration (FDA) requires that all new chemical products be tested. Most of these tests are performed on nonhuman animals. One of the more common tests is the Draize eye irritancy test. The Draize test involves placing a rabbit's head in a restraining device and then putting in one of its eyes a substance such as bleach, shampoo, nail polish, chemical cleaning substances, or weed killer. These experiments are extremely painful. Some rabbits snap their necks in their frantic attempts to escape. In protest, some abolitionist groups have engaged in illegal actions, freeing lab animals and ransacking laboratories. (See Case Study 1: *Animal Liberation in the Science Lab*.)

The medical field, the defense industry, and universities also make extensive use of nonhuman animals in their research. The different types of animal experiments are outlined in the box below. The biggest increase has been in biotechnology and genetic engineering. It may not be long before pigs, genetically altered to contain human genes, will be mass produced as drug factories and to provide organ transplants (**xenotransplantation**) for humans. (See Case Study 2: *Using Animals for Xenotransplants*.) Those who defend the status quo, such as Carl Cohen, argue that animals have no rights that we are bound to respect. They also point out the benefits that animal experimentation has brought to humans.

On the other hand, the results of experiments on one species do not necessarily carry over to humans. Thalidomide, which was responsible for so many devastating birth defects, was tested on animals and deemed safe before being released on the market. In 2006, six healthy male volunteers in a drug trial developed extreme adverse reactions within hours of taking a new drug that had been developed to treat chronic inflammatory conditions and leukemia. This drug also had been tested extensively on nonhuman animals without any adverse reactions.

Millions of dollars are wasted every year on pointless or repetitive experiments. Eliminating these types of experiments would rule out some animal experimentation, but it would not rule out all of it. Both abolitionists and reformists encourage the use of alternatives to animal421research. These include observation of patients, clinical tests, tissue and cell cultures, the use of cadavers, mechanical models, and computer-generated models. Some college students also refuse to participate in dissection and the use of animals in experiments in class.[19]

EXPERIMENTS USING NONHUMAN ANIMALS[18]

§ **Acute toxicity test.** This test involves force-feeding enormous quantities of a substance to a

§ **Acute inhalation toxicity test.** Animals are subjected to large amounts of spray from aerosol preparations. They are then killed and their tissues examined.

§ **Psychology research.** Animals are used in experiments that include electric shock, "punishment," induced fighting and killing, brain damage, mutilation, drug addiction, maternal deprivation, and overcrowding.

§ **Weapons tests.** Animals are used to study the effects of atomic blasts, radiation, chemical warfare, and laser weapons on the body.

§ **Biotechnology.** Animals are genetically engineered for use in medical experiments, to produce drugs for human use, and for organ transplants.

Although some scientists believe that the advent of biotechnology will increase the demand for animal experimentation, others argue that the completion of the Human Genome Project may render much of animal experimentation obsolete. The new field of pharmacogenetics, for example, which studies the effects of genetics on an individual's reactions to drugs, will increase the likelihood that medications will be tailored for each individual based on his or her particular genome rather than on animal models.[20]

THE MORAL ISSUES

The Moral Standing of Nonhuman Animals and Nature

To attribute moral standing to a being is to claim that it is worthy of moral respect and protection. A common approach for determining moral standing is to ask if a being has cognitive qualities similar to those of a rational human adult. The more the being is like a human adult in this respect, the higher its moral standing. Applying this line of reasoning, traditional philosophers and scientists maintain that other animals lack reason and, therefore, have little, if any, moral standing. Carl Cohen represents this view. Peter Singer, on the other hand, argues that some adult mammals are sufficiently like humans in their cognitive abilities to warrant moral consideration. (See readings by Cohen and Singer at the end of this chapter.)

One of the problems with this approach is in knowing the mental life and intentions, if any, of other animals. Furthermore, although lack of certain cognitive abilities may exclude a being from being a moral agent, it does not logically follow that it should be denied moral standing. Regan, for example, argues that nonhuman animals may not be moral agents, but they can still be moral patients; that is, they have interests, and therefore rights, that we ought to respect. For example, a cat companion may not have any moral duties toward us, but we, as moral agents, have a duty to provide her with food and shelter. (See Case Study 3: *The Abandoned Cat*.)

Utilitarians, such as Singer, maintain that sentience, not reason, is the relevant criterion for moral standing. It is wrong to cause suffering no matter what the cognitive level of the being. Singer employs anthropocentric criteria, however, attributing higher moral standing to those animals, such as adult mammals, that are most like adult humans in their cognitive abilities. Buddhist philosophers, on the other hand, cast the net wide enough to include all living beings in the moral

Social Contract Theory

According to social contract theorists, morality is a type of voluntary contract among people. We have moral obligations only toward those who have entered into this agreement. Because nonhuman animals, plants, and inanimate objects such as mountains cannot enter into a social422contract, we have no direct moral obligations toward them. We ought not to kill someone else's pet, not because the pet has rights, but because we have a duty toward the human not to destroy her property. A pet owner, however, may decide to rid herself of her pet.

The Principle of Utility

What matters, according to utilitarians, is whether animals can feel pain, not whether they can reason. Singer, in his writings, graphically illustrates the pain and torment caused to animals by human practices. The moral duty to minimize pain and maximize pleasure militates against most human use of animals. For example, the pleasure humans may get from the taste of meat does not outweigh the suffering caused to animals by farming.

We also need to weigh the benefits to humans of animal experimentation against the animals' interests in living a pain-free life. If our concern is to benefit humans, then it could be argued that it would be preferable to use brain-damaged humans rather than nonhuman animals, because the results would be more accurate. Indeed, the use of certain "nonproductive" groups of humans, such as elderly people and children who are mentally retarded, has been justified on utilitarian grounds by researchers in the past. Most people find this implication of utilitarian reasoning to be morally repugnant.

One problem with relying on utilitarian criteria is the definition of "necessary suffering." The concepts of necessary and unnecessary suffering, used by Cohen to justify the use of nonhuman animals in experiments, are notoriously vague as well as biased in favor of human interests. People who torture their dogs are seen as despicable cowards; scientists who conduct painful experiments on dogs are seen as promoting human progress. Thus, talk of benefits and necessary suffering often serves to mask a view of other animals as merely property. The tremendous power of scientists and agribusiness also ensures that the interests of humans will always be given greater weight than the interests of other animals. The Buddhist principle of *ahimsa*, in contrast, states that the suffering of nonhuman animals cannot be justified by its benefits to humans.

Many environmental ethicists, including the deep ecologists and ecofeminists, reject the utilitarian model. They believe that utilitarians define the "common good" too narrowly and instead embrace a concept of the common good that includes all of nature, not just sentient beings. To use the words of Leopold, "a thing is right when it tends to preserve the stability, integrity, and beauty of the biotic community." The **stewardship** model, like Singer's utilitarian theory, is hierarchical, placing more value on beings who are the most sentient.

Rights

Animal-rights advocates argue that just as we ought to respect humans' intrinsic worth, regardless of their utility, we should also treat other animals as ends-in-themselves. Animal welfarists, while

In contrast, the model of rights adopted by animal-rights advocates, such as Regan, is based on *interests*. The existence of interests is based on the capacity for suffering and for enjoyment. All sentient animals, including humans, have an interest in doing that which brings them pleasure, as well as an interest in avoiding harm and suffering. Under this model of rights, benefits to oneself and others are morally acceptable only if no one else's rights have been violated in achieving these benefits.

Justice, Speciesism, and the Principle of Equality

The **principle of equality** states that it is unjust to treat beings differently unless we can show that there is a difference between them that is relevant to the differential treatment. Singer claims that "**speciesism**," which he defines as "a prejudice or attitude of bias in favor of the interests of members of one's own species and against those of members of another species,"[21] violates the principle of equality. His rule of thumb is that "we should give the same respect to the lives of animals as we give to the lives of those humans at a similar mental level." Both Singer and Warren relate discrimination against other animals to discrimination based on racism and sexism.

The moral equality of sentient beings does not entail that human and nonhuman animals have the same rights. Different species have different interests. There are distinctly human rights, such as the right to religious freedom and the right to a formal education, that other animals lack because they have no interest in either organized religion or formal schooling. All sentient animals, however —including humans, mice, and frogs—have an interest in not being tortured or held captive, not because they are capable of rational thought, but because they have the ability to feel pain.

To most people, speciesism doesn't seem as bad as racism or sexism. In fact, many people find the comparison offensive. Whereas gender or skin color is arbitrary and of no moral importance, they argue, the difference in our treatment of humans and other species is based on morally relevant differences. Only humans, as rational autonomous beings, are able to participate in the moral community. As Singer points out, there are humans who are neither rational nor autonomous. Including these people in the moral community while excluding other animals of equal or greater cognitive capacity violates the principle of equality because it bases moral treatment on group membership rather than on individual differences. Also, there would be no point in using nonhuman animals, such as monkeys and rats, in learning experiments if they were incapable of reason. Indeed, the reason why we use other animals in learning and medical experiments is because they are so much like us.

Natural Law

Defenders of meat-eating point out that other animals eat meat. The human practice of meat-eating is simply part of the natural order. This argument is also used in support of hunting. Opponents of meat-eating maintain that humans are not physiologically suited for a meat-based diet. Indeed, many of our modern ailments are due to our meat-eating habits. This argument also commits the naturalist fallacy: Just because humans can and do eat meat does not mean that they ought to.

requires that we be able to justify actions that affect others. Human beings have the power to exploit or to live in harmony with other species and the environment. It is up to each of us to decide how we want to use this power.

SUMMARY OF READINGS ON ANIMAL RIGHTS AND ENVIRONMENTAL ETHICS

Singer, "Animal Liberation." Animals are sentient beings that deserve our moral respect. Therefore, meat-eating and most animal experimentation are immoral.

Cohen, "Do Animals Have Rights?" Animals do not have moral rights because they are incapable of asserting them.

Garvey, "Climate Changes and Moral Outrage." Both individuals and states have an obligation to reduce global warming.

Gore, "Testimony on Climate Change." We need to find the moral courage to resolve the current climate crisis.

PETER SINGER

Animal Liberation

Australian philosopher Peter Singer is a professor of bioethics at the University Center for Human Values, Princeton University, and laureate professor at the Centre for Applied Philosophy and Public Ethics at the University of Melbourne in Australia. Singer argues that utilitarian theory requires that the interests of all sentient beings be given equal weight. To not take the pain and interests of other animals seriously is to engage in what Singer calls "speciesism."

Critical Reading Questions

1. What is the principle of equality?

2. What is "speciesism"? Why is speciesism wrong, according to Singer?

3. According to Singer, what kind of lives have greater moral value than others?

4. Why does Singer claim that most humans are speciesists? What does he mean when he says that speciesism is analogous to racism and sexism?425

5. What is the moral significance of sentience, or the ability to feel pain? In what ways do humans

9. Why do psychological experiments that use animals pose a dilemma for researchers who claim that other animals are not like us cognitively?

10.According to Singer, when, if ever, is animal experimentation morally justified?

11.Why does Singer claim that the controversy over the benefits derived from animal experimentation is essentially irrelevant?

Peter Singer, "All Animals Are Equal" from *Animal Liberation: The Definitive Classic of the Animal Movement* (Harper Perennial Modern Classics, 2009). Used with permission of the author. Notes have been omitted.

Jeremy Bentham, the founder of the reforming utilitarian school of moral philosophy, incorporated the essential basis of moral equality into his system of ethics by means of the formula: "Each to count for one and none for more than one." In other words, the interests of every being affected by an action are to be taken into account and given the same weight as the like interests of any other being….

It is an implication of this principle of equality that our concern for others and our readiness to consider their interests ought not to depend on what they are like or on what abilities they may possess. Precisely what our concern or consideration requires us to do may vary according to the characteristics of those affected by what we do: concern for the well-being of children growing up in America would require that we teach them to read; concern for the well-being of pigs may require no more than that we leave them with other pigs in a place where there is adequate food and room to run freely. But the basic element—the taking into account of the interests of the being, whatever those interests may be—must, according to the principle of equality, be extended to all beings, black or white, masculine or feminine, human or nonhuman….

It is on this basis that the case against racism and the case against sexism must both ultimately rest; and it is in accordance with this principle that the attitude that we may call "speciesism," by analogy with racism, must also be condemned. Speciesism—the word is not an attractive one, but I can think of no better term—is a prejudice or attitude of bias in favor of the interests of members of one's own species and against those of members of other species. It should be obvious that the fundamental objections to racism and sexism made by Thomas Jefferson and Sojourner Truth apply equally to speciesism. If possessing a higher degree of intelligence does not entitle one human to use another for his or her own ends, how can it entitle humans to exploit nonhumans for the same purpose?

Many philosophers and other writers have proposed the principle of equal consideration of interests, in some form or other, as a basic moral principle; but not many of them have recognized that this principle applies to members of other species as well as to our own. Jeremy Bentham was one of the few who did realize this. In a forward-looking passage written at the time when black slaves had been freed by the French but in the British dominions were still being treated in the way we now treat animals, Bentham wrote:

The day *may* come when the rest of the animal creation may acquire those rights which never could

In this passage Bentham points to the capacity for suffering as the vital characteristic that gives a being the right to equal consideration. The capacity for suffering—or more strictly, for suffering and/or enjoyment or happiness—is not just another characteristic like the capacity for language or higher mathematics. Bentham is not saying that those who try to mark "the insuperable line" that determines whether the interests of a being should be considered happen to have chosen the wrong characteristic. By saying that we must consider the interests of all beings with the capacity for suffering or enjoyment Bentham does not arbitrarily exclude from consideration any interests at all—as those who draw the line with reference to the possession of reason or language do. The capacity for suffering and enjoyment is *a prerequisite for having interests at all*, a condition that must be satisfied before we can speak of interests in a meaningful way. It would be nonsense to say that it was not in the interests of a stone to be kicked along the road by a schoolboy. A stone does not have interests because it cannot suffer. Nothing that we can do to it could possibly make any difference to its welfare. The capacity for suffering and enjoyment is, however, not only necessary, but also sufficient for us to say that a being has interests—at an absolute minimum, an interest in not suffering. A mouse, for example, does have an interest in not being kicked along the road, because it will suffer if it is….

Racists violate the principle of equality by giving greater weight to the interests of members of their own race when there is a clash between their interests and the interests of those of another race. Sexists violate the principle of equality by favoring the interests of their own sex. Similarly, speciesists allow the interests of their own species to override the greater interests of members of other species. The pattern is identical in each case.

Most human beings are speciesists…. Ordinary human beings—not a few exceptionally cruel or heartless humans, but the overwhelming majority of humans—take an active part in, acquiesce in, and allow their taxes to pay for practices that require the sacrifice of the most important interests of members of other species in order to promote the most trivial interests of our own species….

Do animals other than humans feel pain? How do we know? Well, how do we know if anyone, human or nonhuman, feels pain? We know that we ourselves can feel pain. We know this from the direct experience of pain that we have when, for instance, somebody presses a lighted cigarette against the back of our hand. But how do we know that anyone else feels pain? We cannot directly experience anyone else's pain, whether that "anyone" is our best friend or a stray dog. Pain is a state of consciousness, a "mental event," and as such it can never be observed. Behavior like writhing, screaming, or drawing one's hand away from the lighted cigarette is not pain itself; nor are the recordings a neurologist might make of activity within the brain observations of pain itself. Pain is something that we feel, and we can only infer that others are feeling it from various external indications.

In theory, we *could* always be mistaken when we assume that other human beings feel pain. It is conceivable that one of our close friends is really a cleverly constructed robot, controlled by a brilliant scientist so as to give all the signs of feeling pain, but really no more sensitive than any other machine. We can never know, with absolute certainty, that this is not the case. But while this might present a puzzle for philosophers, none of us has the slightest real doubt that our close

Nearly all the external signs that lead us to infer pain in other humans can be seen in other species, especially the species most closely related to us—the species of427mammals and birds. The behavioral signs include writhing, facial contortions, moaning, yelping or other forms of calling, attempts to avoid the source of pain, appearance of fear at the prospect of its repetition, and so on. In addition, we know that these animals have nervous systems very like ours, which respond physiologically as ours do when the animal is in circumstances in which we would feel pain: an initial rise of blood pressure, dilated pupils, perspiration, an increased pulse rate, and, if the stimulus continues, a fall in blood pressure. Although human beings have a more developed cerebral cortex than other animals, this part of the brain is concerned with thinking functions rather than with basic impulses, emotions, and feelings. These impulses, emotions, and feelings are located in the diencephalon, which is well developed in many other species of animals, especially mammals and birds.

We also know that the nervous systems of other animals were not artificially constructed—as a robot might be artificially constructed—to mimic the pain behavior of humans. The nervous systems of animals evolved as our own did, and in fact the evolutionary history of human beings and other animals, especially mammals, did not diverge until the central features of our nervous systems were already in existence. A capacity to feel pain obviously enhances a species' prospects of survival, since it causes members of the species to avoid sources of injury. It is surely unreasonable to suppose that nervous systems that are virtually identical physiologically, have a common origin and a common evolutionary function, and result in similar forms of behavior in similar circumstances should actually operate in an entirely different manner on the level of subjective feelings….

Other differences between humans and animals cause other complications. Normal adult human beings have mental capacities that will, in certain circumstances, lead them to suffer more than animals would in the same circumstances. If, for instance, we decided to perform extremely painful or lethal scientific experiments on normal adult humans, kidnapped at random from public parks for this purpose, adults who enjoy strolling in parks would become fearful that they would be kidnapped. The resultant terror would be a form of suffering additional to the pain of the experiment. The same experiments performed on non-human animals would cause less suffering since the animals would not have the anticipatory dread of being kidnapped and experimented upon. This does not mean, of course, that it would be *right* to perform the experiment on animals, but only that there is a reason, which is *not* speciesist, for preferring to use animals rather than normal adult human beings, if the experiment is to be done at all. It should be noted, however, that this same argument gives us a reason for preferring to use human infants—orphans perhaps—or severely retarded human beings for experiments, rather than adults, since infants and retarded humans would also have no idea of what was going to happen to them. So far as this argument is concerned nonhuman animals and infants and retarded humans are in the same category; and if we use this argument to justify experiments on nonhuman animals we have to ask ourselves whether we are also prepared to allow experiments on human infants and retarded adults; and if we make a distinction between animals and these humans, on what basis can we do it, other than a bare-faced —and morally indefensible—preference for members of our own species?

It may be objected that comparisons of the sufferings of different species are impossible to make and that for this reason when the interests of animals and humans clash the principle of equality gives no guidance. It is probably true that comparisons of suffering between members of different species cannot be made precisely, but precision is not essential. Even if we were to prevent the infliction of suffering on animals only when it is quite certain that the interests of humans will not be428affected to anything like the extent that animals are affected, we would be forced to make radical changes in our treatment of animals that would involve our diet, the farming methods we use, experimental procedures in many fields of science, our approach to wildlife and to hunting, trapping and the wearing of furs, and areas of entertainment like circuses, rodeos, and zoos. As a result, a vast amount of suffering would be avoided….

Just as most human beings are speciesists in their readiness to cause pain to animals when they would not cause a similar pain to humans for the same reason, so most human beings are speciesists in their readiness to kill other animals when they would not kill human beings….

This does not mean that to avoid speciesism we must hold that it is as wrong to kill a dog as it is to kill a human being in full possession of his or her faculties. The only position that is irredeemably speciesist is the one that tries to make the boundary of the right to life run exactly parallel to the boundary of our own species. Those who hold the sanctity of life view do this, because while distinguishing sharply between human beings and other animals they allow no distinctions to be made within our own species, objecting to the killing of the severely retarded and the hopelessly senile as strongly as they object to the killing of normal adults.

To avoid speciesism we must allow that beings who are similar in all relevant respects have a similar right to life—and mere membership in our own biological species cannot be a morally relevant criterion for this right. Within these limits we could still hold, for instance, that it is worse to kill a normal adult human, with a capacity for self-awareness and the ability to plan for the future and have meaningful relations with others, than it is to kill a mouse, which presumably does not share all of these characteristics; or we might appeal to the close family and other personal ties that humans have but mice do not have to the same degree; or we might think that it is the consequences for other humans, who will be put in fear for their own lives, that makes the crucial difference; or we might think it is some combination of these factors, or other factors altogether.

Whatever criteria we choose, however, we will have to admit that they do not follow precisely the boundary of our own species. We may legitimately hold that there are some features of certain beings that make their lives more valuable than those of other beings; but there will surely be some non-human animals whose lives, by any standards, are more valuable than the lives of some humans. A chimpanzee, dog, or pig, for instance, will have a higher degree of self-awareness and a greater capacity for meaningful relations with others than a severely retarded infant or someone in a state of advanced senility. So if we base the right to life on these characteristics we must grant these animals a right to life as good as, or better than, such retarded or senile humans.

This argument cuts both ways. It could be taken as showing that chimpanzees, dogs, and pigs, along with some other species, have a right to life and we commit a grave moral offense whenever

fact that a being is a member of our own species is not in itself enough to make it always wrong to kill that being, we may come to reconsider our policy of preserving human lives at all costs, even when there is no prospect of a meaningful life or of existence without terrible pain.

I conclude, then, that rejection of speciesism does not imply that all lives are of equal worth. While self-awareness, the capacity to think ahead and have hopes and aspirations for the future, the capacity for meaningful relations with others and so on are not relevant to the question of inflicting pain—since pain is pain, whatever other capacities, beyond the capacity to feel pain, the being may have—these capacities are relevant to the question of taking life. It is not arbitrary to hold that the life of a self-aware being, capable of abstract thought, of planning for the future, of complex acts of communication, and so on, is more valuable than the life of a being without these capacities….

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The practice of experimenting on nonhuman animals as it exists today throughout the world reveals the consequences of speciesism. Many experiments inflict severe pain without the remotest prospect of significant benefits for human beings or any other animals. Such experiments are not isolated instances, but part of a major industry. In Britain, where experimenters are required to report the number of "scientific procedures" performed on animals, official government figures show that 3.5 million scientific procedures were performed on animals in 1988. In the United States there are no figures of comparable accuracy….

Among the tens of millions of experiments performed, only a few can possibly be regarded as contributing to important medical research. Huge numbers of animals are used in university departments such as forestry and psychology; many more are used for commercial purposes, to test new cosmetics, shampoos, food coloring agents, and other inessential items. All this can happen only because of our prejudice against taking seriously the suffering of a being who is not a member of our own species. Typically, defenders of experiments on animals do not deny that animals suffer. They cannot deny the animals' suffering, because they need to stress the similarities between humans and other animals in order to claim that their experiments may have some relevance for human purposes. The experimenter who forces rats to choose between starvation and electric shock to see if they develop ulcers (which they do) does so because the rat has a nervous system very similar to a human being's, and presumably feels an electric shock in a similar way.

There has been opposition to experimenting on animals for a long time. This opposition has made little headway because experimenters, backed by commercial firms that profit by supplying laboratory animals and equipment, have been able to convince legislators and the public that opposition comes from uninformed fanatics who consider the interests of animals more important than the interests of human beings. But to be opposed to what is going on now it is not necessary to insist that all animal experiments stop immediately. All we need to say is that experiments serving no direct and urgent purpose should stop immediately, and in the remaining fields of research, we should, whenever possible, seek to replace experiments that involve animals with alternative methods that do not….

This attitude is illustrated by the following autobiographical statement [that] appeared in *New*

Perhaps not many psychologists would now proudly state that their work has nothing to do with the human mind. Nevertheless many of the experiments that are performed on rats can only be explained by assuming that the experimenters really are interested in the behavior of the rat for its own sake, without any thought of learning anything about humans. In that case, though, what possible justification can there be for the infliction of so much suffering? It is certainly not for the benefit of the rat.

So the researcher's central dilemma exists in an especially acute form in psychology: either the animal is not like us, in which case there is no reason for performing the experiment; or else the animal is like us, in which case we ought not to perform on the animal an experiment that would be considered outrageous if performed on one of us….

Once a pattern of animal experimentation becomes the accepted mode of research in a particular field, the process is self-reinforcing and difficult to break out of. Not only publications and promotions but also the awards and grants that finance research become geared to animal experiments. A proposal for a new experiment with animals is something that the administrators of research funds will be ready to support, if they have in the past supported other experiments on animals. New methods that do not make use of animals will seem less familiar and will be less likely to receive support.

All this helps to explain why it is not always easy for people outside the universities to understand the rationale for the research carried out under university auspices. Originally, perhaps, scholars and researchers just430set out to solve the most important problems and did not allow themselves to be influenced by other considerations. No doubt some are still motivated by these concerns. Too often, though, academic research gets bogged down in petty and insignificant details because the big questions have been studied already and they have either been solved or proven too difficult. So the researchers turn away from the well-plowed fields in search of new territory where whatever they find will be new, although the connection with a major problem may be remote. It is not uncommon, as we have seen, for experimenters to admit that similar experiments have been done many times before, but without this or that minor variation; and the most common ending to a scientific publication is "further research is necessary." …

When are experiments on animals justifiable? Upon learning of the nature of many of the experiments carried out, some people react by saying that all experiments on animals should be prohibited immediately. But if we make our demands as absolute as this, the experimenters have a ready reply: Would we be prepared to let thousands of humans die if they could be saved by a single experiment on a single animal?

This question is, of course, purely hypothetical. There has never been and never could be a single experiment that saved thousands of lives. The way to reply to this hypothetical question is to pose another: Would the experimenters be prepared to carry out their experiment on a human orphan under six months old if that were the only way to save thousands of lives?

If the experimenters would not be prepared to use a human infant then their readiness to use nonhuman animals reveals an unjustifiable form of discrimination on the basis of species, since

So far as we know, human infants possess no morally relevant characteristic to a higher degree than adult nonhuman animals, unless we are to count the infants' potential as a characteristic that makes it wrong to experiment on them. Whether this characteristic should count is controversial—if we count it, we shall have to condemn abortion along with experiments on infants, since the potential of the infant and the fetus is the same. To avoid the complexities of this issue, however, we can alter our original question a little and assume that the infant is one with irreversible brain damage so severe as to rule out any mental development beyond the level of a six-month-old infant. There are, unfortunately, many such human beings, locked away in special wards throughout the country, some of them long since abandoned by their parents and other relatives, and, sadly, sometimes unloved by anyone else. Despite their mental deficiencies, the anatomy and physiology of these infants are in nearly all respects identical with those of normal humans. If, therefore, we were to force-feed them with large quantities of floor polish or drip concentrated solutions of cosmetics into their eyes, we would have a much more reliable indication of the safety of these products for humans than we now get by attempting to extrapolate the results of tests on a variety of other species. The LD50 tests, the Draize eye tests, the radiation experiments, the heatstroke experiments, and many others could have told us more about human reactions to the experimental situation if they had been carried out on severely brain-damaged humans instead of dogs or rabbits.

So whenever experimenters claim that their experiments are important enough to justify the use of animals, we should ask them whether they would be prepared to use a brain-damaged human being at a similar mental level to the animals they are planning to use. I cannot imagine that anyone would seriously propose carrying out the experiments described in this chapter on brain-damaged human beings. Occasionally it has become known that medical experiments have been performed on human beings without their consent; one case did concern institutionalized intellectually disabled children, who were given hepatitis. When such harmful experiments on human beings become known, they usually lead to an outcry against the experimenters, and rightly so. They are, very often, a further example of the arrogance of the research worker who justifies everything on the grounds of increasing knowledge. But if the experimenter claims that the experiment is important431enough to justify inflicting suffering on animals, why is it not important enough to justify inflicting suffering on humans at the same mental level? What difference is there between the two? Only that one is a member of our species and the other is not? But to appeal to that difference is to reveal a bias no more defensible than racism or any other form of arbitrary discrimination….

No doubt there are some fields of scientific research that will be hampered by any genuine consideration of the interests of animals used in experimentation. No doubt there have been some advances in knowledge which would not have been attained as easily without using animals. Examples of important discoveries often mentioned by those defending animal experimentation go back as far as Harvey's work on the circulation of blood. They include Banting and Best's discovery of insulin and its role in diabetes; the recognition of poliomyelitis as a virus and the development of a vaccine for it; several discoveries that served to make open heart surgery and coronary artery bypass graft surgery possible; and the understanding of our immune system and ways to overcome rejection of transplanted organs. The claim that animal experimentation was essential in making these discoveries has been denied by some opponents of experimentation. I do not intend to go into

many false leads would also not have been pursued, and it is possible that medicine would have developed in a very different and more efficacious direction, emphasizing healthy living rather than cure.

In any case, the ethical question of the justifiability of animal experimentation cannot be settled by pointing to its benefits for us, no matter how persuasive the evidence in favor of such benefits may be. The ethical principle of equal consideration of interests will rule out some means of obtaining knowledge. There is nothing sacred about the right to pursue knowledge. We already accept many restrictions on scientific enterprise. We do not believe that scientists have a general right to perform painful or lethal experiments on human beings without their consent, although there are many cases in which such experiments would advance knowledge far more rapidly than any other method. Now we need to broaden the scope of this existing restriction on scientific research.

Finally, it is important to realize that the major health problems of the world largely continue to exist, not because we do not know how to prevent disease and keep people healthy, but because no one is putting enough effort and money into doing what we already know how to do. The diseases that ravage Asia, Africa, Latin America, and the pockets of poverty in the industrialized West are diseases that, by and large, we know how to cure. They have been eliminated in communities that have adequate nutrition, sanitation, and health care….

It does not seem likely that any major Western democracy is going to abolish all animal experimentation at a stroke. Governments just do not work like that. Animal experimentation will only be ended when a series of piecemeal reforms have reduced its importance, led to its replacement in many fields, and largely changed the public attitude to animals. The immediate task, then, is to work for these partial goals, which can be seen as milestones on the long march to the elimination of all exploitation of sentient animals. All concerned to end animal suffering can try to make known what is happening at universities and commercial laboratories in their own communities. Consumers can refuse to purchase products that have been tested on animals—especially in cosmetics, alternatives are now available. Students should decline to carry out experiments they consider unethical. Anyone can study the academic journals to find out where painful experiments are being carried out, and then find some way of making the public aware of what is happening….

The exploitation of laboratory animals is part of the larger problem of speciesism and it is unlikely to be eliminated altogether until speciesism itself is eliminated. Surely one day, though, our children's children, reading about what was done in laboratories in the twentieth century, will feel the same sense of horror and incredulity at what otherwise civilized people could do that we now feel when we read about the atrocities of the Roman gladiatorial arenas or the eighteenth-century slave trade.

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Discussion Questions

1. Discuss Singer's claim that most humans are speciesists. If you are a speciesist, on what grounds do you morally justify it? In what ways does your lifestyle contribute to the suffering of the

4. Like Singer, former philosophy professor R. G. Frey opposed species membership as the criterion of moral standing.[22] Whereas Singer concludes that the principle of equality should preclude experimentation on nonhuman animals whose cognitive capacity is equivalent to that of human infants or cognitively impaired humans, Frey drew the opposite conclusion. Humans with an impoverished cognitive life can be used in experiments in the same way we use animals. Discuss Frey's argument as well as how Singer might respond to him.

5. Is Singer, as a utilitarian, being inconsistent when he argues that the benefits derived from animal experimentation are morally irrelevant? Do the benefits of experimentation ever justify the use of nonconsenting humans or nonhuman animals in experiments? Support your answers.

CARL COHEN

Do Animals Have Rights?

Carl Cohen is a philosophy professor at the University of Michigan. Cohen defends the traditional Kantian view of moral agency and rights. According to Cohen, rights are based on self-assertion rather than on interests. Because only moral agents can assert moral claims, only moral agents have rights. Therefore, we do not violate the rights of nonhuman animals by doing research on them.

Critical Reading Questions

1. Why is the question of whether animals have rights of great importance?

2. How does Cohen define the term *right?*

3. Why does Cohen reject Regan's argument that animals have rights?

4. According to Cohen, what is the relationship between rights and obligations?

5. Why can't nonhuman animals be the bearers of rights? How does Cohen use the example of the baby zebra and the lioness to illustrate his position?433

6. How does Cohen interpret Regan's concept of nonhuman animals as being moral patients?

7. According to Cohen, where do human rights come from?

8. How does Cohen respond to the objection that, under his definition, human infants and other cognitively impaired humans don't have rights?

9. What fallacy does Cohen accuse Regan of committing in his use of the term *inherent value?* According to Cohen, how does this fallacy invalidate Regan's conclusion that animals have rights?

You have a right to the return of money you lent me; we both understand that. It may be very convenient for me to keep the money, and you may have no need of it whatever; but my convenience and your needs are not to the point. You have a *right* to it, and we have courts of law partly to ensure that such rights will be respected….

A great deal was learned about hypothermia by some Nazi doctors who advanced their learning by soaking Jews in cold water and putting them in refrigerators to learn how hypothermia proceeds. We have no difficulty in seeing that they may not advance medicine in that way; the subjects of those atrocious experiments had rights that demanded respect. For those who ignored their rights we have nothing but moral loathing.

Some persons believe that animals have rights as surely as those Jews had rights, and they therefore look on the uses of animals in medical investigations just as we look at the Nazi use of the Jews, with moral loathing. They are consistent in doing so. If animals have rights they certainly have the right not to be killed, even to advance our important interests.

Some may say, "Well, they have rights, but we have rights too, and our rights override theirs." That may be true in some cases, but it will not solve the problem because, although we may have a weighty *interest* in learning, say, how to vaccinate against polio or other diseases, we do not have a *right* to learn such things. Nor could we honestly claim that we kill research animals in self-defense; they did not attack us. If animals have rights, they certainly have the right not to be killed to advance the interests of others, whatever rights those others may have.

In 1952 there were about 58,000 cases of polio reported in the United States, and 3,000 polio deaths; my parents, parents everywhere, trembled in fear for their children at camp or away from home. Polio vaccination became routine in 1955, and cases dropped to about a dozen a year; today polio has been eradicated completely from the Western Hemisphere. The vaccine that achieved this, partly developed and tested only blocks from where I live in Ann Arbor, could have been developed *only* with the substantial use of animals. Polio vaccines had been tried many times earlier, but from those earlier vaccines children had contracted the disease; investigators had become, understandably, exceedingly cautious.

The killer disease for which a vaccine now is needed most desperately is malaria, which kills about 2 million people each year, most of them children. Many vaccines have been tried—not on children, thank God—and have failed. But very recently, after decades of effort, we learned how to make a vaccine that does, with complete success, inoculate mice against malaria. A safe vaccine for humans we do not yet have—but soon we will have it, thanks to the use of those mice, many of whom will have died in the process. To test that vaccine first on children would be an outrage…. We use mice or monkeys434*because there is no other way* … to determine the reliability and safety of new vaccines without repeated tests on live organisms. Therefore, because we certainly may not use human children to test them, we will use mice (or as we develop an AIDS vaccine, primates) *or we will never have such vaccines*.

But if those animals we use in such tests have rights as human children do, what we did and are doing to them is as profoundly wrong as what the Nazis did to those Jews not long ago. Defenders

The harms others might face as a result of the dissolution of [some] practice or institution is no defense of allowing it to continue…. No one has a right to be protected against being harmed if the protection in question involves violating the rights of others…. No one has a right to be protected by the continuation of an unjust practice, one that violates the rights of others…. Justice *must* be done, though the … heavens fall. (pp. 346–347)

That last line echoes Kant, who borrowed it from an older tradition. Believing that rats have rights as humans do, Regan (1983) was convinced that killing them in medical research was morally intolerable. He wrote,

On the rights view, [he means, of course, the Regan rights view] we cannot justify harming a single rat *merely* by aggregating "the many human and humane benefits" that flow from doing it.

… Not even a single rat is to be treated as if that animal's value were reducible to his *possible utility* relative to the interests of others. (p. 384)

If there are some things that we cannot learn because animals have rights, well, as Regan (1983) put it, so be it.

This is the conclusion to which one certainly is driven if one holds that animals have rights. If Regan is correct about the moral standing of rats, we humans can have no right, ever, to kill them— unless perchance a rat attacks a person or a human baby, as rats sometimes do; then our right of self-defense may enter, I suppose. But medical investigations cannot honestly be described as self-defense, and medical investigations commonly require that many mice and rats be killed. Therefore, all medical investigations relying on them, or any other animal subjects—which includes most studies and all the most important studies of certain kinds—will have to stop….

WHY ANIMALS DO NOT HAVE RIGHTS

Many obligations are owed by humans to animals; few will deny that. But it certainly does not follow from this that animals have rights because it is certainly not true that every obligation of ours arises from the rights of another. Not at all. We need to be clear and careful here. Rights entail obligations. If you have a right to the return of the money I borrowed, I have an obligation to repay it. No issue. If we have the right to speak freely on public policy matters, the community has the obligation to respect our right to do so. But the proposition *all rights entail obligations* does not convert simply, as the logicians say. From the true proposition that all trees are plants, it does not follow that all plants are trees. Similarly, not all obligations are entailed by rights. Some obligations, like mine to repay the money I borrowed from you, do arise out of rights. But many obligations are owed to persons or other beings who have no rights whatever in the matter.

Obligations may arise from commitments freely made: As a college professor I accept the obligation to comment at length on the papers my students submit, and I do so; but they have not the right to *demand* that I do so….

Special relations often give rise to obligations: Hosts have the obligation to be cordial to their guests, but the guest has not the right to demand cordiality. Shepherds have obligations to their

… Some of our most important obligations—to members of our family, to the needy, to neighbors, and to sentient creatures of every sort—have no foundation in rights at all. Correlativity appears critical from the perspective of one who holds a right; your right correlates with my obligation to respect it. But the claim that rights and obligations are *reciprocals*, that *every* obligation flows from another's right, is false, plainly inconsistent with our general understanding of the differences between what we think we *ought* to do, and what others can justly *demand* that we do.

I emphasize this because, although animals have no rights, it surely does not follow from this that one is free to treat them with callous disregard. Animals are not stones; they feel. A rat may suffer; surely we have the obligation not to torture it gratuitously, even though it be true that the concept of a right could not possibly apply to it. We humans are obliged to act humanely, that is, being aware of their sentience, to apply to animals the moral principles that govern us regarding the gratuitous imposition of pain and suffering; which is not, of course, to treat animals as the possessors of rights.

Animals cannot be the bearers of rights because the concept of rights is essentially *human;* it is rooted in, and has force within, a human moral world. Humans must deal with rats—all too frequently in some parts of the world—and must be moral in their dealing with them; but a rat can no more be said to have rights than a table can be said to have ambition. To say of a rat that it has rights is to confuse categories, to apply to its world a moral category that has content only in the human moral world.

Try this thought experiment. Imagine, on the Serengeti Plain in East Africa, a lioness hunting for her cubs. A baby zebra, momentarily left unattended by its mother, is the prey; the lioness snatches it, rips open its throat, tears out chunks of its flesh, and departs. The mother zebra is driven nearly out of her wits when she cannot locate her baby; finding its carcass she will not even leave the remains for days. The scene may be thought unpleasant, but it is entirely natural, of course, and extremely common. If the zebra has a right to live, if the prey is just but the predator unjust, we ought to intervene, if we can, on behalf of right. But we do not intervene, of course—as we surely would intervene if we saw the lioness about to attack an unprotected human baby or you. What accounts for the moral difference? We justify different responses to humans and to zebras on the ground (implicit or explicit) that their moral stature is very different. The human has a right not to be eaten alive; it is, after all, a human being. Do you believe the baby zebra has the *right* not to be slaughtered by that lioness? That the lioness has the *right* to kill that baby zebra for her cubs? If you are inclined to say, confronted by such natural rapacity—duplicated with untold variety millions of times each day on planet earth—that neither is right or wrong, that neither has a *right* against the other, I am on your side. Rights are of the highest moral consequence, yes; but zebras and lions and rats are totally amoral; there is no morality for them; they do no wrong, ever. In their world there are no rights.

A contemporary philosopher who has thought a good deal about animals, referring to them as "moral patients," put it this way:

A moral patient lacks the ability to formulate, let alone bring to bear, moral principles in deliberating

But humans do have rights. Where do our rights come from? Why are we not crudely natural creatures like rats and zebras? …

To be a moral agent (on this view) is to be able to grasp the generality of moral restrictions on our will. Humans understand that some things, which may be in436our interest, *must not be willed;* we lay down moral laws for ourselves, and thus exhibit, as no other animal can exhibit, moral autonomy. My dog knows that there are certain things she must not do—but she knows this only as the outcome of her learning about her interests, the pains she may suffer if she does what had been taught forbidden. She does not know, cannot know (as Regan agrees) that any conduct is wrong. The proposition *It would be highly advantageous to act in such-and-such a way, but I may not because it would be wrong* is one that no dog or mouse or rabbit, however sweet and endearing, however loyal or attentive to its young, can ever entertain, or intend, or begin to grasp. Right is not in their world. But right and wrong are the very stuff of human moral life, the ever-present awareness of human beings who can do wrong, and who by seeking (often) to avoid wrong conduct prove themselves members of a moral community in which rights may be exercised and must be respected.

Some respond by saying, "This can't be correct, for human infants (and the comatose and senile, etc.) surely have rights, but they make no moral claims or judgments and can make none—and any view entailing that children can have no rights must be absurd." Objections of this kind miss the point badly. It is not individual persons who qualify (or are disqualified) for the possession of rights because of the presence or absence in them of some special capacity, thus resulting in the award of rights to some but not to others. Rights are universally human; they arise in a *human moral world*, in a moral *sphere*. In the human world moral judgments are pervasive; it is the fact that all humans including infants and the senile are members of that moral community—not the fact that as individuals they have or do not have certain special capacities, or merits—that makes humans bearers of rights. Therefore, it is beside the point to insist that animals have remarkable capacities, that they really have a consciousness of self, or of the future, or make plans, and so on. And the tired response that because infants plainly cannot make moral claims they must have no rights at all, or rats must have them too, we ought forever put aside. Responses like these arise out of a misconception of right itself. They mistakenly suppose that rights are tied to some identifiable individual abilities or sensibilities, and they fail to see that rights arise only in a community of moral beings, and that therefore there are spheres in which rights do apply and spheres in which they do not.

Rationality is not at issue; the capacity to communicate is not at issue. My dog can reason, if rather weakly, and she certainly can communicate…. Nor is the capacity to suffer here at issue. And, if *autonomy* be understood only as the capacity to choose this course rather than that, autonomy is not to the point either. But *moral autonomy*—that is, *moral self-legislation*—is to the point, because moral autonomy is uniquely human and is for animals out of the question, as we have seen, and as Regan and I agree. In talking about autonomy, therefore, we must be careful and precise….

WHY ANIMALS ARE MISTAKENLY BELIEVED TO HAVE RIGHTS

depending on the obligation to be kind or not to be cruel, and those depending on any kind of utilitarian calculation….

The case is built entirely on the principle that allegedly *carries over* almost everything earlier claimed about human rights to rats and other animals. What principle is that? It is the principle, put in italics but given no name, that equates moral agents with moral patients:

The validity of the claim to respectful treatment, and thus the case for the recognition of the right to such treatment, cannot be any stronger or weaker in the case of moral patients than it is in the case of moral agents. (Regan, p. 279)

But hold on. Why in the world should anyone think this principle to be true? Back where Regan first recounted his view of moral patients, he allowed that some of them are, although capable of experiencing pleasure and pain, lacking in other capacities. But he is interested, he told us437there, in those moral patients—those animals—that are like humans in having *inherent value*. This is the key to the argument for animal rights, the possession of inherent value. How that concept functions in the argument becomes absolutely critical. I will say first briefly what will be shown more carefully later: *Inherent value* is an expression used by Regan (and many like him) with two very different senses—in one of which it is reasonable to conclude that those who have inherent value have rights, and in another sense in which that inference is wholly unwarranted. But the phrase *inherent value* has some plausibility in both contexts, and thus by sliding from one sense of inherent value to the other Regan appears to succeed … in making the case for animal rights.

… Regan went on to argue for the proposition that all moral agents are "equal in inherent value." Holding some such views we are likely to say, with Kant, that all humans are beyond price. Their inherent value gives them moral dignity, a unique role in the moral world, as agents having the capacity to act morally and make moral judgments. This is inherent value in Sense 1.

The expression *inherent value* has another sense, however, also common and also plausible. My dog has inherent value, and so does every wild animal, every lion and zebra, which is why the senseless killing of animals is so repugnant. Each animal is unique, not replaceable in itself by another animal or by any rocks or clay. Animals, like humans, are not just things; they live, and as unique living creatures they have inherent value. This is an important point, and again likely to be thought plausible; but here, in Sense 2, the phrase *inherent value* means something quite distinct from what was meant in its earlier uses.

Inherent value in Sense 1, possessed by all humans but not by all animals, which warrants the claim of human rights, is very different from inherent value in Sense 2, which warrants no such claim. The uniqueness of animals, their intrinsic worthiness as individual living things, does not ground the possession of rights, has nothing to do with the moral condition in which rights arise. Regan's argument reached its critical objective with almost magical speed because, having argued that beings with inherent value (Sense 1) have rights that must be respected, he quickly asserted (putting it in italics lest the reader be inclined to express doubt) that rats and rabbits also have rights because they, too, have inherent value (Sense 2).

of the phrase *inherent value* is obscured (from the author, I believe, as well as from the reader) is the fact that the assertion that animals have rights appears only indirectly, as the outcome of the application of the principle that moral patients are entitled to the same respect as moral agents—a principle introduced at a point in the book long after the important moral differences between moral patients and moral agents have been recognized, with a good deal of tangled philosophical argument having been injected in between….

Animals do not have rights. Right does not apply in their world. We do have many obligations to animals, of course, and I honor Regan's appreciation of their sensitivities…. But he is, I submit, profoundly mistaken. I conclude with the observation that, had his mistaken views about the rights of animals long been accepted, most successful medical therapies recently devised—antibiotics, vaccines, prosthetic devices, and other compounds and instruments on which we now rely for saving and improving human lives and for the protection of our children—could not have been developed; and were his views to become general now (an outcome that is unlikely but possible) the consequences for medical science and for human well-being in the years ahead would be nothing less than catastrophic.

Advances in medicine absolutely require experiments, many of which are dangerous. Dangerous experiments absolutely require living organisms as subjects. Those living organisms (we now agree) certainly may not be human beings. Therefore, most advances in medicine will continue to rely on the use of nonhuman438animals, or they will stop. Regan is free to say in response, as he does, "so be it." The rest of us must ask if the argument he presents is so compelling as to force us to accept that dreadful result.

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Discussion Questions

1. Is Cohen a speciesist? Discuss how Singer might respond to Cohen's claim that all and only humans have rights, even those who lack reason and moral autonomy.

2. Cohen acknowledges that inhumane treatment of nonhuman animals is wrong when we can achieve the same results using alternative methods. Discuss whether his claim that researchers have a moral obligation to treat nonhuman animals humanely is consistent with his claim that animals have no rights.

3. Cohen admits that nonhuman animals feel pain and that researchers should not subject them to unnecessary suffering. What does he mean by unnecessary suffering? Compare and contrast Cohen's and Singer's positions on the significance of suffering in the moral justification of experiments using nonhuman animals.

4. Does Cohen do an effective job of discrediting Regan's argument that nonhuman animals have rights? Support your answer. Discuss how Regan might respond to Cohen's criticism of his position.

5. Discuss how Cohen might respond to Singer's argument that scientists who justify using

Climate Change and Moral Outrage

Critical Reading Questions

1. Why is it easy for morality to "Waver out of focus" when thinking about climate change?

2. On what grounds do many people, including Garvey, believe that the developed world's failure to take action on climate change is a moral wrong?

3. How do our actions affect people in poorer nations?439

4. What is Singer's Principle and how does it apply to our moral obligation regarding climate change?

5. What is Garvey's second principle and how does this increase our obligation as a developed nation to take action?

6. According to Garvey what do the golden rule, Kant's categorical imperative, and Bentham's principle of utility tell us about our moral obligations when it comes to climate change?

7. What moral obligations, according to Garvey, do we have as individuals to modify our life styles, and why?

James Garvey, "Climate Change and Moral Outrage" *Human Ecology Review*, vol. 17, no. 2 (Winter 2010) pp 96–101. Copyright © 2010 Society for Human Ecology. Used with permission of the Society for Human Ecology.

By many accounts, the decisions made by our generation will have profound consequences for the future of our planet and those who come after us. Yet we have done and continue to do little or nothing about our changing climate. The ray of light in all of this has a great deal to do with the moral dimension of climate change. Human beings sometimes change course when they see that what they are doing is unbearably wrong. The ethics of climate change can push us in exactly the right direction.

However, reflection on the ethics of climate change can get us into trouble too. It can get us into philosophical trouble, because it is easy to make mistakes when thinking about rights and wrongs on a planetary scale. Morality, whatever it is, seems to waver out of focus when applied to the big picture. It does not like that sort of thing and feels more comfortable in homey contexts, probably because it grew up in small towns and copes best with little wrongs.

If we catch someone red-handed, right there in front of us, shop-lifting a bottle of tequila, it is easy to come to the conclusion that what is going on is wrong. With climate change, though, causes and effects are spread out in space and time…. Actions set in motion in one hemisphere have effects on the other side of the world. The way the land is used here affects flooding over there. The fuel burned over there changes the El Niño a little, which causes a drought somewhere else. Causes and effects are smeared out in time as well. It takes a while for our actions to translate into noticeable effects on the climate. We are feeling the effects of decisions taken more than a century ago, and what we do now will have effects long after our deaths.

Suppose that a billion little causes, rippling out into a trillion little effects criss-crossing over many years and all over the planet, conspire through a complex causal chain to raise the sea-level in 2111, inundating a coastal village in China, ruining crops and destroying potable water, resulting in the loss of many lives. Probably we can tell that there is harm in there, but whose fault is it? Whose responsibility is it? Who should have done otherwise? Can we really say, with a straight face, that we did it, in our microscopic share of those trillion criss-crossing effects? Can one really see oneself as hooked up to the environment in such a way that one's teeny contribution to this slow-motion, distant disaster constitutes a genuine wrong? Maybe this is exactly what we have to do, if we are to take action on climate change.

Reflection on the ethics of climate change, even on a smaller scale, can get us into other sorts of trouble too. Primarily, it annoys other people. Not only can it end up sounding like moralizing, rather than moral philosophy, but it gets us where we live. It issues in the conclusion that our comfy lives of high-energy consumption have to change, that we in the developed world should make serious sacrifices for other people. Arriving at this conclusion is not very difficult, but seeing it clearly and acting on it certainly is. What gets in the way, partly, are certain consequentialist worries, particularly problems440associated with seeing our small role in the temporally and spatially and causally spread-out confusion that is the problem of climate change. The aim of this paper is to get past those worries or at least head in the right direction. We will start by homing in on the West's moral failings in this connection.

A LACK OF MORAL RECTITUDE

Many people believe that the developed world's failure to take action on climate change amounts to a moral wrong, perhaps an enormous moral wrong…. The right spokespeople for this view might really be those presently on the receiving end of some of the worst effects of climate change. Antigua and Barbuda is one of the small island states, like Tuvalu and the Carteret Islands, whose existence is threatened by the predicted rise in sea levels owed to climate change. Its ambassador to the United States, Lionel Hurst, gave a speech in 2002 at the International Red Cross Conference on Climate Change and Natural Disasters. He said a great deal, but consider just these lines: 'We see a lack of moral rectitude by those who are in leadership positions, who know the consequences of their inaction, and yet insist that they will not act…. [the] thirst for environmental justice must be cast in moral terms…. It must be seen as good versus evil' (Hurst, 2002).

That is strong stuff—particularly when you realize that the leaders he has in mind are ours, the ones running the West. We are where the evil is….

Rather a lot has been said about the previous U.S. administration's efforts to deny or play down or conceal worrying findings about the future of our planet. Still more has been said about American efforts to wreck the Kyoto Protocol as well as recent efforts to muddle things in Bali. The developed world's biggest polluters fought hard to take the teeth out of Kyoto. Members of the EU managed to negotiate the right to club together as a single entity for the purposes of counting carbon emissions, secure in the knowledge that plenty of wiggle room would be made by recent members whose emissions were dropping as fast as their economies. Some argue that things in the West are

behaviour is a kind of moral outrage, I want to spring another conclusion on you. It is a conclusion which might matter, anyway a conclusion which might undermine what is the single most common personal excuse for failing to take individual action on climate change. It has something to do with our earlier inability to keep a straight face while entertaining the thought that our teeny contribution to climate change is morally wrong…. We'll start, though, with an argument for the claim that the West's failure to take action on climate change is morally wrong.

FACTS AND SINGER'S PRINCIPLE

A lot of people accept the fact that the present state of play is somehow unjust or wrong. You can arrive at this conclusion in just a few paragraphs. Burning fossil fuels thickens the blanket of greenhouse gasses around our world, and the world warms up as a result. The warmer our planet becomes, the more suffering we are in for—suffering caused by failed crops, hotter days and nights, rising sea levels, dwindling water supplies, altered patterns of disease, conflict over shifting resources, and more dramatic weather. This connection between fossil fuels and suffering has a lot to do with the fact that our planet's carbon sinks cannot absorb all of our emissions. The sinks are therefore a limited and valuable resource.

Some countries on the planet—the richer, more developed, industrialised ones—have used up more than a fair share of the sinks and therefore caused more of the suffering which is underway and on the cards. If one thinks a little about fairness or justice or responsibility for441harm, or the importance of doing something about unnecessary human suffering, then one will quickly be drawn to the conclusion that the rich countries have a moral obligation to reduce emissions. Maybe they should pay for a few sea walls in Bangladesh, possibly foot the bill for a bit of disaster relief, too. Zoom in on the thought that the developed world has a moral obligation to reduce its emissions. Its failure to do so is tied to human suffering. The fact that the world's polluters have not taken meaningful action is an obvious wrong. It seems easy enough to see it.

Singer, who is better at this than I am, only needs two sentences to make essentially the same point. His first sentence presents a fact, and the second offers a moral, interpretive principle which leads to a conclusion about action on climate change: 'To put it in terms a child could understand, as far as the atmosphere is concerned, the developed nations broke it. If we believe that people should contribute to fixing something in proportion to their responsibility for breaking it, then the developed nations owe it to the rest of the world to fix the problem with the atmosphere' (Singer, 2004)….

The USA, with less than 5% of the world's population, is responsible for an enormous share of carbon dioxide emissions by country each year: about 20% of the global total. The European Union is responsible for much more than half of this, almost 14% of the global total. The numbers then drop off pretty quickly, with Russia and India each responsible for about 5% of the global total. China recently overtook the US and now emits a bit more than 20% of the world's total emissions. Try to bear in mind, as you think about this, that China has about a billion more people in it than the United States.

behaviour of the developed world is morally wrong. If you think for a moment about the people who suffer now and those who will suffer as a result, the West's behaviour might strike you as a moral outrage.

MORE FACTS AND A SECOND PRINCIPLE

Thinking a little about room for reduction and capacity for reduction can make a second principle clear, a principle having to do with being able but unwilling to do the right thing. Consider room for eduction first.

Not all emissions are morally equivalent. Some emissions might be quantitatively identical but differ dramatically in moral quality. The greenhouse gasses resulting from a long-haul flight for a weekend break on some sandy beach are not on a par with an equal quantity of emissions resulting from the efforts of subsistence farmers toiling away in a field. As Shue puts it, some emissions are luxury emissions and others are subsistence emissions, and if cuts must be made, it is the former which have to go first (Shue, 1993). It nearly goes without saying that the West emits considerably more luxury emissions than the developing world, and it therefore has more room for reduction.

Think now about the capacity for reduction, the varying abilities of states to make cuts in emissions or otherwise shift resources around. It is fairly obvious that the West is best placed to make large cuts in a number of senses. The developed world has the strength to move mountains. Its people are formally educated for longer, and the technological options available to them are greater. Compared to the poor countries of the world, the rich nations have better infrastructures, a greater capacity to produce and store food, better healthcare, better housing, more manpower, more money, and on and on and on.

The developed world has not just the room for reduction, but also the capacity and the resources generally to do what is right. The developed world is best-placed for442action on climate change by just about any measure you like. The fact that it has done so little when it is most able to take action is grounds for a second argument for the conclusion that its behaviour is a moral outrage.

Take a moment to think about the facts and principles underpinning this conclusion. The developed world is primarily responsible for a problem with our atmosphere. Singer's principle tells us that there's a connection between damaging something and an obligation to fix it. The developed world has done the most damage to our planet, and it continues to use a disproportionate share of our planet's carbon sinks. It, therefore, has the largest responsibility to take serious action on climate change.

The developed world also has the room and the capacity to take the necessary action, certainly as compared to the developing world. The fact that it fails to do so, against this background, is another reason to think that the developed world is doing something wrong in its failure to take action on climate change. There is a principle behind this conclusion too: the greater the ability to do what is right, the greater the obligation to do what is right. Like Singer's principle, this one is uncontroversial. One would have some explaining to do if one walked past a drowning child and did nothing to help. One would have a lot more explaining to do if one were a physically fit and well-

Consistency is near the heart of reflection on moral matters…. The ancient injunction, 'do unto others as you would have them do unto you', just is the call for the consistency of principle in human relationships. Kant's reflections on the universalizability of maxims put consistency at the centre of things. Consistency lies just behind Bentham's insistence that we give equal consideration to the pleasure and pain of all interested parties when deciding what to do—it is not just our pain or our friend's pain that counts.

… All things being roughly equal, if I think some moral judgement applies to you in such and such circumstances I've got no grounds to complain about that judgement applying to me if I find myself in those circumstances too. If I think stealing is wrong when you do it, consistency demands that I've got to admit that stealing is wrong when I do it, too.

If the thoughts scouted above about climate change lead you to the conclusion that the behaviour of the West is a moral outrage, then consistency of principle might lead you to the conclusion that your own behaviour is a moral outrage too. Think again about Singer's principle and those thoughts about the lifeguard. Couple those two principles with a new set of facts—facts not about the West, but about you. If it is correct to think that the US and other countries in the West are wrong to do nothing meaningful about climate change despite being responsible for the largest share of emissions, then it is correct to think that we are wrong to do nothing in our everyday lives despite being responsible for the largest emissions per capita. People who live in the US are responsible for nearly 20 metric tons of carbon dioxide on average each year. Australians are responsible for 18 metric tons. Canadians emit a little more than 17 metric tons. People in many EU countries—like the Netherlands, the UK and Germany—emit around 10 metric tons on average….

Residents of more than half of the countries on our planet, including the Chinese, emit less than 5 metric tons each year. People living in India are responsible for just over 1 metric ton each year. Residents of more than a third of the countries on the planet are responsible for less than even a single metric ton. Many human beings are responsible for no measurable emissions at all. Compared to most people on the planet, the greenhouse gas emissions resulting from our individual lives in the West are enormous.

If it is correct to think that the West does wrong by doing nothing despite having the room to reduce emissions and the capacity to do so, then it is correct to think that we are doing wrong in our everyday lives too. Plenty of your emissions are luxury emissions; most do not result from securing the real necessities of life. Probably, also, you have advantages when it comes to taking action on climate change as compared to many people on the planet, and those advantages line up with the443ones we considered a moment ago when thinking about the developed world. We have plenty of cash to spare, certainly as compared to others on our world, and not just the desperately poor. Probably we are well-placed to take action in other ways too, just given the fact that we live in the relative safety of a developed country: we are healthy and wellfed, we have easy access to the information required to do what is right, we are not trapped in a refugee camp, there are no snipers about, we can avail ourselves of energy-saving lifestyle choices, and we can express ourselves freely and push for a greener world.

matter all that much—so why bother changing them? … next to nothing in our history has prepared us for careful thinking about a spatially smeared out, causally jumbled, intergenerational slow-motion disaster. Thinking our way through it will require new thoughts, but we can make some progress with the morality we already have. A large part of that is built on consistency.

Does talk of consistency just side-step consequentialism? Reflection on consequences has plenty to do with the conclusions we just reached about the developed world. The trouble is that consequentialism is not much help when we try to think about our moral connection to the smeared out effects and causes of climate change. The demand for consistency can help put teeth on that old bit of neo-hippy wisdom: think globally, act locally. It can help us avoid various moral mistakes having to do with hypocrisy, too.

There is another sort of concern which sometimes surfaces in this connection. It has to do with the behaviour of other countries, particularly China. Maybe it's hard to portray us in the West as the bad guys when we're no longer at the top of the emissions list. Shouldn't China make some cuts too? The usual thing to do here is to point to complications concerning cumulative and per capita emissions—we're still the villains in several morally relevant senses. There are also reasonable points to make about our obligation to help developing countries leapfrog into green energy rather than point the finger. After all, we've effectively clogged up the planet's carbon sinks and thereby blocked the cheapest path to progress. But I want to insist on a different answer: the moral demands placed on us in the West are what they are no matter what China does. You don't get to lie just because other people do. At any rate dredging up realpolitik in the middle of a conversation about ethics can only muddy the waters.

That said, there is still a fair and live moral question here. Suppose China should make some cuts, and the only way to leverage them into doing the right thing is for us to hold out too. Should we put our moral obligations to one side for a moment in pursuit of some greater good? Should we wait in an effort to secure a binding deal that has the best effects overall? These are questions partly about how we rank what matters to us. Notice, however, that in order to ask them you have to accept the moral demands placed on us, and that's all I was after in the first place. Once we have a grip on those demands, it's possible to come to further conclusions about real and no doubt difficult political questions—do as much realpolitik as you like. What we mustn't do, though, is allow the murky political questions to obscure the clear moral case for action. We ought to try things the other way around—get a grip on what's right first, and then find the political means to achieve it.

As time goes on, though, the moral situation is likely to shift around. Those responsible and those affected will not neatly distribute themselves into rich and poor, East and West, North and South forever. There are complications already, and as the present century grinds on, the bulk of the responsibility for the damage to our world could well be owed to the Chinese, the Indians, or even a mishmash of international corporations or individuals or … who knows? If the facts change we'll need to think about all of this again, but given the facts that we've got and the principles we've accepted, the444conclusion for us and for the West is clear enough: we ought to take meaningful action right now.

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Discussion Questions

1. Critically analyze Garvey's claim that we as a developed nation have a greater obligation to take action against global warming.

3. Is your life a "moral outrage" and, if so, what you going to do about it? Examine your own lifestyle and future plans in light of Garvey's argument that we as individual bear some moral responsibility for global warming.

4. Discuss how Al Gore would most likely respond to Garvey's arguments as well as what policies they might propose for dealing with climate change.

AL GORE

Testimony before U.S. House of Representatives on Global Warming and the Climate Crisis

In his testimony, former Vice President, Congressman, and longtime environmentalist Al Gore outlines some of the steps already taken to address climate changes. While these steps are good, he concludes that more has to be done and that we need to find the moral courage to resolve the current climate crisis.

Critical Reading Questions

1. What or who, according to Gore, is the primary cause of global warming?

2. What evidence does Gore provide to support his claim that global warming is occurring?

3. What are some of the steps taken so far to counter global warming?

4. Why does Gore claim that more needs to be done?

5. What are our moral obligations as individuals to take action?

I want to testify today about what I believe is a planetary emergency—a crisis that threatens the survival of our civilization and the habitability of the Earth. Just six weeks ago, the scientific community, in its strongest statement to date, confirmed that the evidence of warming is "unequivocal." Global warming is real and human activity is the main cause. The consequences are mainly negative and headed toward catastrophic, unless we act. However, the good news is that we can meet this challenge. It is not too late, and we have everything we need to get started.

Testimony of the Honorable Al Gore before the U.S. House of Representatives Energy & Commerce Committee Subcommittee on Energy & Air Quality and the Science & Technology Committee Subcommittee on Energy & Environment. March 21, 2007.

As many know, the Chinese expression for "crisis" consists of two characters side by side. The first symbol means "danger." The second symbol means "opportunity."446I would like to discuss both the danger and the opportunity here today.

First of all, there is no longer any serious debate over the basic points that make up the consensus

carbon dioxide and there are billions of tons underneath the permafrost.

However, there is a great deal of new momentum for action to solve the climate crisis. Today, I am here to deliver more than a half million messages to Congress asking for real action on global warming. More than 420 Mayors have now adopted Kyoto-style commitments in their cities and have urged strong federal action. The evangelical and faith communities have begun to take the lead, calling for measures to protect God's creation. The State of California, under a Republican Governor and a Democratic legislature, passed strong, economy wide legislation mandating cuts in carbon dioxide. Twenty-two states and the District of Columbia have passed renewable energy standards for the electricity sector. Much more needs to be done, but change is in the air.

I do not believe that the climate crisis should be a partisan political issue. I just returned from the United Kingdom, where last week the two major parties put forward their climate change platforms. The Tory and Labour parties are in vigorous competition with one another—competing to put forward the best solution to the climate crisis. I look forward to the day when we return to this way of thinking here in the U.S.

The climate crisis is, by its nature, a global problem—and ultimately the solution must be global as well. The best way - and the only way - to get China and India on board is for the U.S. to demonstrate real leadership. As the world's largest economy and greatest superpower, we are uniquely situated to tackle a problem of this magnitude.

After all, we have taken on problems of this scope before. When England and then America and our allies rose to meet the threat of global Fascism, together we won two wars simultaneously in Europe and the Pacific.

This is a moral moment of similar magnitude. This is not ultimately about any scientific discussion or political dialogue. It is about who we are as human beings and our capacity to transcend our limitations and rise to meet this challenge.

The solutions to this problem are accessible, but politically - at least in the near term - seem quite difficult. In practice, however, they will turn out to be much easier than they appear to us now.

For example, the Montreal Protocol on Substances that Deplete the Ozone Layer first negotiated in the 1980's was opposed by industry for fear it would hurt the economy because its provisions were too stringent. However, governments and industry rose to meet the challenge and the treaty was strengthened twice in quick succession to quickly ramp down the chemicals that were causing the hole in the ozone layer.

There are some who will say that acting to solve this crisis will be costly. I don't agree. If we solve it in the right way, we will save money and boost productivity. Moreover, the consequences of inaction would be devastating to both the environment and the economy. Recent reports make that clear.

When I think about the climate crisis today I can imagine a time in the future when our children and grandchildren ask us one of two questions. Either they will ask: What were you thinking, didn't you care about our future? Or they will ask: How did you find the moral courage to cross party lines and

with Trump and Gore. Construct an argument and solution that takes into consideration, as much as possible, both their concerns.

2. There is a divide in the environmental movement over whether the environment has rights. Ecocentrists argue that it does have rights while anthropocentrists maintain that the reason for preserving the environment are based on human interests. Which view seems closest to that expressed by Gore? Discuss how our view of the moral value of the environment, as well as that of traditional Western philosophers, has contributed to the current climate change crisis.

3. In his book *Earth in the Balance* (1992), Al Gore argues that modern civilization is dysfunctional because it assumes a radical separation between humans and nature. This dysfunction manifested our addiction to consuming ever larger quantities of oil, trees, topsoil, and other "substances we rip from the crust of the earth." At the same time, Gore rejects deep ecology, claiming that it incorrectly portrays humans in the "role of pathogens, a kind of virus giving the earth a rash and a fever, threatening the planet's vital life functions."[23] Gore argues instead that we need to seek a balance between our needs and preservation of the global environment. Do you agree with Gore? Support your answer. Discuss also how James Garvey might respond to Gore.

4. Gore maintains that each of us needs to find the moral courage to choose a future that will help resolve the climate change crisis. Discuss your career and lifestyle choices in light of Gore's challenge.

CASE STUDIES

1. ANIMAL LIBERATION IN THE SCIENCE LAB

The Animal Liberation Front (ALF) is a loose organization of radical animal-rights activists in over forty countries who target science laboratories, slaughterhouses, and the fur and lumber industries. Since its founding in England in 1976, the ALF engaged in hundreds, if not thousands, of reported direct actions. One of the most publicized actions in the United States took place in 1984, when five members of the ALF broke into the Experimental Head Injury Lab at the University of Pennsylvania and stole files and videotapes of experiments. The videotapes showed gruesome scenes of terrified baboons in vises with their heads being smashed by pistons while the researchers joked around. The tapes showed operations being performed on primates without regard for their pain or for standard research procedures. After taking the videotapes, the ALF ransacked the lab.

In the controversy that followed the release of the tapes to the public, Dr. Thomas Gennarelli, the director of the lab, defended the research, claiming that the animals had been properly treated. He also accused the ALF of setting back medical research. Both the university and the National Institutes of Health (NIH), which gave the lab a new grant to repair the damage, supported Dr. Gennarelli.

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The ALF responded to the accusation by comparing the lab experiments to those conducted by Nazi Dr. Joseph Mengele on Jews in concentration camps. Protesters supported the ALF by staging

Discussion Questions

1. The AFL has a policy of nonviolence toward living beings, including people engaged in animal experimentation. Were stealing the tapes and ransacking the lab morally justified? Support your answers. Discuss how a utilitarian, such as Peter Singer, might respond to the tactics used by the AFL.

2. Discuss how an abolitionist, a reformist, and a status quo person would respond to this case. What solutions would each of them most likely propose?

3. Discuss whether the experiments would have been morally justified if Dr. Gennarelli had not violated the Animal Welfare Act and if it could be shown that thousands of human lives have been saved as a result of these experiments.

4. Is the AFL a terrorist organization? See Chapter 9, page 371, for a definition of "terrorism." Support your answer.

5. The Animal Rights Direct Action Coalition has targeted McDonald's restaurants. They drive up to a McDonald's in a pickup truck with a dead cow in the back and a sign reading "Here's your lunch." Discuss the moral implications of these tactics.

2. USING ANIMALS FOR XENOTRANSPLANTS

More than 100,000 people a year in the United States are waiting for organ transplants, according to the United Network for Organ Sharing, which oversees organ donations. Most of these people will die before a suitable donor is found. Xenotransplants, the transplanting of organs from one species of animal into another, may offer a solution to this shortage.

The first animal-to-human transplant occurred in 1906, when French physician Mathieu Jaboulay transplanted a kidney from a pig into a woman. Neither the woman nor the pig survived the procedure. Since then several people have received organs from pigs and baboons. In 2001 the Secretary's Advisory Committee on Xenotransplantation (SACX) was formed to advise the U.S. Department of Health and Human Services. Since other animals, such as pigs, carry viruses that are harmless to them but might be lethal to humans, one concern is the risk of inadvertently introducing an infectious disease via a xenotransplant.

Although baboons are genetically closer to humans, many researchers prefer pigs for xenotransplants because pigs are anatomically very similar to humans. They are also healthier, less likely to carry viruses, and easier to breed. Scientists are genetically engineering pigs with human genes so that their organs will be more compatible with humans. To minimize the risk of porcine449viruses, cells and organs used for xenotransplantation are harvested from pigs raised under pathogen-free conditions.

Discussion Questions

1. Although many people have ethical qualms about using baboons for xenotransplants because of their similarity to humans, they have fewer objections to using pigs because pigs are already

endanger healthy human populations, should they be banned or more closely regulated? Support your answers.

4. Weigh the rights, if any, of nonhuman animals such as pigs and baboons against those of humans who need organ transplants. Discuss whether it is morally acceptable to sacrifice the lives of other animals for humans whose organs have been damaged by their destructive lifestyles, such as heavy drinking or smoking. Discuss how Cohen and Singer might respond to this question.

3. THE ABANDONED CAT[26]

David had always wanted a cat. Although pets were forbidden in his dormitory, shortly after he moved to college David went to the local animal shelter and adopted a young cat, which he kept confined in his room. At the end of the year, David moved back home and left the cat to fend for itself.

After several weeks of wandering around campus, the cat was taken in by one of the department secretaries, who took pity on it. By this time the cat was near starvation; it also had a rash that had caused it to lose much of its fur. The secretary knew that one of the professors in the department, Professor Carey, was thinking of getting a cat. Professor Carey agreed to take the cat home.

When the cat's rash did not clear up, Professor Carey took it to the veterinarian. The vet told the professor that the cat had multiple allergies and would have to receive cortisone shots as well as eat a special diet. In addition, the cat had an overactive thyroid that would require an expensive operation or else medication for the rest of its life. The medication would cost approximately $30 a month and would have to be given to the cat three times a day at six-hour intervals. After thinking about it, Professor Carey decided that the expense and inconvenience were not worth it. She asked the veterinarian to euthanize the cat.

Discussion Questions

1. What are the professor's moral obligations, if any, toward the cat? Is it morally acceptable for her to euthanize the cat? Or does the cat's interest in staying alive outweigh the professor's inconvenience? Are there any conditions under which it is morally acceptable to have an450animal companion euthanized? Support your answers. Discuss how Cohen and Singer might respond to this question.

2. Discuss whether it was morally acceptable for David to adopt a cat from the animal shelter, knowing that it was against the rules to keep a cat in the dormitory and knowing that he could not take the cat home with him once the school year ended.

3. Some animal-rights activists argue that owning house pets (animals that are kept inside or allowed outside only on a leash or in an enclosure) is a type of slavery. Do you agree? Discuss whether having a pet or animal companion can be compatible with respect.

4. A lot of animal shelters do not allow college students to adopt cats and dogs. Is this policy a violation of the liberty rights of college students? Support your answer.

In 1993 it was announced that the London Zoo would have to close down. A high-profile "Save Our Zoo" campaign, however, raised enough money to keep the zoo going. Supporters of the zoo argued that zoos are often the first contact young people have with "wild" animals. They also noted the important role that the London Zoo plays in educating people about conservation and in saving endangered species.

Opponents respond that watching captive and anguished animals is hardly a good introduction. The use of documentaries of animals in their natural habitat is a more realistic way to educate young people. They also argue that conservation, including the preservation of endangered species, should take place within the animals' natural habitats.

Discussion Questions

1. Are zoos ever morally justified? If so, under what conditions? What about safari parks, such as Six Flags or Disney's Animal Kingdom, where animals roam free, within limits, and the people remain in their cars? Support your answers.

2. While it may be possible to protect some of these species, such as the black rhino, within their natural habitats, in other cases the habitats have been destroyed by human encroachment. The destruction of the rain forests in Central and South America for agricultural land has already resulted in the extinction of hundreds of species of animals. Discuss whether in such cases the continuation of zoos is morally justified as a means of preserving endangered species.

5. THE "BAMBI BOOM"

The Teneja family were on the way to the College of William and Mary where nineteen-year-old Baninder Teneja was starting a summer research project. As they exited Interstate 95 on a side trip to nearby Lake Anna, a panicked deer struck and careened off a van ahead of them and came crashing through their front window. Baninder, who was riding in the back seat, was fatally451injured when she was struck by the decapitated head of the deer before it smashed through their back window.

This was just one of an estimated 1.5 million deer/vehicle collisions that occur in the United States each year, causing more than 150 human fatalities. Vehicle/deer accidents are causing increasing concern as the deer population in the United States continues to grow at an unprecedented rate. In many parts of the United States, the deer population is two to four times what it was in pre-European times. This population explosion is due to a number of factors, including the milder winters of the last decade and a decline in the number of natural predators such as the wolf.

Humans aren't the only ones being negatively impacted by the "Bambi boom." Some scientists predict that the deer population, if left unchecked, could lead to the catastrophic disintegration of certain biotic communities. Overbrowsing by deer not only damages their own habitat but reduces vegetation that butterflies and songbirds use. Their eating habits are also threatening plant species such as orchids and lilies as well as eastern hemlock and white cedar trees.

Discussion Questions

3. Aldo Leopold supported the "harvesting" of deer herds to control their population. If what is right is defined in terms of preserving the "integrity, stability and beauty of the biotic community," critically analyze whether the land ethic also justify the culling of human populations where overpopulation is threatening the integrity of the biotic environment. Discuss how Al Gore might respond to this question.

4. Working in small groups, develop an ethical public policy for dealing with the Bambi boom.

6. EARTH'S DWINDLING FORESTS

People, like other animals, need nature in order to survive. We use trees for lumber to build our homes, the soil for growing crops, the earth as a source of minerals and energy. However, are there limits to how much we ought to use and, if so, what is the justification for these limits?

Most of the world's forests have been cleared by humans, including about 20 percent of the world's rain forests.[27] Tropical rain forests have been bulldozed to create land for cattle grazing, mining operations, hydroelectric dams, and the cultivation of export crops such as coffee. In addition to destruction of species and wildlife habitat, the lifestyles as well as the lives of indigenous peoples have been threatened by the clearing operations and by epidemics such as measles.

The rain forests are not the only forests that are being destroyed at an alarming rate. In the United States, the U.S. Forest Service has long regarded assisting the logging industry as one of its primary roles. There is little wilderness left in our national forests that is not crisscrossed with logging roads. In some cases the logging has threatened bird species with extinction and destroyed recreational areas, creating animosity between environmentalists and the businesses452and workers whose livelihood depends on harvesting natural resources. Deforestation also reduces rainfall and contributes to global warming.

Discussion Questions

1. Discuss the prevailing attitude and practices regarding using natural resources in light of the ecofeminists' concept of the logic of dominance. What policies might Garvey suggest for the human use of rain forests and our national forests?

2. Much of the coffee North Americans drink is grown on land that used to be rain forests. This is because coffee plants grown on bare land are much more productive than coffee plants that are shade-grown. Discuss whether we have a moral obligation either to give up coffee drinking or to drink only shade-grown coffee, even though it is more expensive and harder to find in stores.

3. One of the primary reasons for clearing rain forests is to raise beef for the U.S. market. Do we have a moral obligation to modify our lifestyle in order to use fewer natural resources? If so, what is the source of this moral obligation? Discuss how Al Gore might answer this question.

7. THE GULF OIL SPILL DISASTER

In April 2010 the BP Deepwater Horizon oil rig exploded killing eleven workers and spewing four million barrels of light crude oil into the Gulf of Mexico. It was the largest oil spill in U.S. history and

compensate victims of the spill. Disputes over compensation are still ongoing.

Discussion Questions

1. BP claims it has already compensated commercial fishermen "far in excess of the documented, spill-related losses," noting that 2011 was a banner year for tourism and fishing.[28] Residents of Louisiana and the Gulf coast respond that regional economic recovery is still uneven. Discuss to what extent people affected by a disaster have a right to continued compensation; or whether they have a duty to seek other means of livelihood.

2. President Obama opened up the Arctic waters for offshore oil drilling over protests from environmental groups who fear a similar oil spill disaster. Environmentalists argue that it would be difficult or almost impossible to contain an oil spill in the Artic because of the remoteness of the area, the ice cover, and the lack of daylight during the winter months. Using the utilitarian calculus, and taking into consideration our need for oil and the concerns of environmentalists, discuss whether we should be drilling for oil in the Arctic.453

3. President Trump wants to open up the Arctic National Wildlife Refuge for oil drilling. Supporters of this plan point out the economic advantages, including less dependency on foreign markets for oil and the creation of new jobs. Opponents argue that drilling in the refuge could have serious consequences for the ecosphere, including the risk of pollution from an oil spill and the large-scale displacement and destruction of wildlife and fish. Which side do you think makes the best argument, and why? Discuss also how both Al Gore and a utilitarian might respond to this controversy.

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4. Aristotle, "Politics," bk. 1, ch. 8, in *The Oxford Translation of Aristotle*, trans. Benjamin Jowett (Oxford: Oxford University Press).

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7. Charles Darwin, *The Descent of Man*, excerpt from Tom Regan and Peter Singer, eds., *Animal Rights and Human Obligations* (Englewood Cliffs, N.J.: Prentice Hall, 1989), 30.

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23. Al Gore, *Earth in the Balance: Ecology and the Human Spirit* (Boston: Houghton Mifflin, 1992), 216, 221.

24. Terry Frieden, "FBI, ATF Address Domestic Terrorism," May 19, 2005, CNN.com.

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26. This is based on a true story. However, the professor in question—the author—did not have the cat euthanized.

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